

PS04_Solution_ClaudeAI

ClaudeAI solution

ECEN 4/5632 — Problem Set 4 Solutions

Problem 1: Difference Equation to $H(z)$

Given: $y[n] - 1.4y[n-1] + 0.45y[n-2] = x[n] + x[n-1]$

a) Find $H(z)$

Taking the Z-transform of both sides (assuming zero initial conditions):

$$Y(z)[1 - 1.4z^{-1} + 0.45z^{-2}] = X(z)[1 + z^{-1}]$$

$$H(z) = \frac{1 + z^{-1}}{1 - 1.4z^{-1} + 0.45z^{-2}}$$

b) Zeros and poles

Multiply numerator and denominator by z^2 :

$$H(z) = \frac{z^2 + z}{z^2 - 1.4z + 0.45} = \frac{z(z + 1)}{z^2 - 1.4z + 0.45}$$

Zeros: $z = 0$ and $z = -1$

Poles: solve $z^2 - 1.4z + 0.45 = 0$

$$z = [1.4 \pm \sqrt{(1.96 - 1.8)}]/2 = [1.4 \pm 0.4]/2$$

Poles: $z = 0.9$ and $z = 0.5$

c) ROC for the causal system

For a causal system, the ROC is the region *outside* the pole of largest magnitude:

ROC: $|z| > 0.9$

d) BIBO stability

A causal LTI system is BIBO stable iff the ROC includes the unit circle. Since both poles (0.9 and 0.5) lie strictly inside the unit circle, the causal ROC $|z| > 0.9$ contains $|z| = 1$.

Yes — the system is BIBO stable.

Problem 2: Geometric Frequency-Response Reasoning

Given: $H(z) = (1 + z^{-1})/(1 - 0.8z^{-1})$

a) Poles and zeros

Writing $H(z) = (z + 1)/(z - 0.8)$:

- **Zero:** $z = -1$
- **Pole:** $z = 0.8$

b) Geometric evaluation

For $z = e^{j\omega}$ on the unit circle:

$$|H(e^{j\omega})| = \frac{|e^{j\omega} - (-1)|}{|e^{j\omega} - 0.8|} = \frac{\text{distance to zero}}{\text{distance to pole}}$$

At $\omega = 0$ ($z = 1$):

- Distance to zero at -1 : $|1 - (-1)| = 2$ (long vector — evaluation point is far from the zero)
- Distance to pole at 0.8 : $|1 - 0.8| = 0.2$ (very short vector — evaluation point is right next to the pole)
- $|H(e^{j0})| = 2 / 0.2 = \mathbf{10}$

At $\omega = \pi$ ($z = -1$):

- Distance to zero at -1 : $|-1 - (-1)| = 0$ (the evaluation point *sits on top of* the zero)
- Distance to pole at 0.8 : $|-1 - 0.8| = 1.8$
- $|H(e^{j\pi})| = 0 / 1.8 = \mathbf{0}$

Sketch description: Draw the unit circle with the pole marked as an $\times$ at $z = 0.8$ (on the positive real axis, inside the circle) and the zero marked as an $\circ$ at $z = -1$ (on the circle itself, at the leftmost point). At $\omega = 0$, the point $z = 1$ is close to the pole and far from the zero $\rightarrow$ short pole-vector, long zero-vector $\rightarrow$ large gain. At $\omega = \pi$, the point $z = -1$ coincides exactly with the zero $\rightarrow$ zero-vector length is 0 $\rightarrow$ the response is completely nulled.

c) Conclusion

The gain is large near $\omega = 0$ and drops to exactly zero at $\omega = \pi$. This is a **lowpass filter**: it passes low frequencies (and even boosts DC) while completely rejecting the Nyquist frequency.

Problem 3: 600 Hz Notch Filter Design ($f_s = 9000$ Hz)

a) Pole/zero locations

$$\omega_0 = \frac{2\pi f_0}{f_s} = \frac{2\pi(600)}{9000} = \frac{2\pi}{15} \approx 0.4189 \text{ rad} \approx 24^\circ$$

Zeros (on the unit circle, radius 1, at $\pm\omega_0$):

$$z = e^{\pm j\omega_0} = \cos(24^\circ) \pm j\sin(24^\circ) \approx 0.9135 \pm j 0.4067$$

Poles (radius $r = 0.96$, same angle):

$$z = 0.96 e^{\pm j\omega_0} \approx 0.8770 \pm j 0.3904$$

This gives the standard second-order notch transfer function:

$$H(z) = \frac{1 - 2\cos(\omega_0)z^{-1} + z^{-2}}{1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2}}$$

b) Effect of increasing r from 0.96 to 0.99

Moving the poles closer to the unit circle (closer to the zeros) **narrows the notch bandwidth**, making the null much sharper/more selective around exactly 600 Hz while leaving frequencies just outside the notch nearly unaffected. The trade-off: the impulse response rings longer (slower pole decay, since the pole radius is closer to 1), and the filter becomes more sensitive to coefficient quantization/rounding errors — a small perturbation in r or ω_0 has a larger relative effect on the notch shape when r is very close to 1.

Problem 4: Ideal Lowpass FIR Filter

a) Inverse DTFT

$$h_d[n] = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \frac{\sin(\omega_c n)}{\pi n}, \quad n \neq 0$$
$$h_d[0] = \frac{\omega_c}{\pi}$$

(This can also be written as $h_d[n] = \frac{\omega_c}{\pi} \text{sinc}\left(\frac{\omega_c n}{\pi}\right)$ using the normalized sinc, $\text{sinc}(x) = \sin(\pi x)/(\pi x)$.)

b) Sketch for $\omega_c = \pi/3$, $n \in [-8, 8]$

$$h_d[0] = \frac{\pi/3}{\pi} = \frac{1}{3} \approx 0.3333$$

Selected values (h_d is even, $h_d[-n] = h_d[n]$):

n	0	± 1	± 2	± 3	± 4	± 5	± 6	± 7	± 8
$h_d[n]$	0.33	0.2757	0.1378	0.0000	-0.068	-0.055	0.0000	0.0394	0.0345
	33				9	1			

Sketch description: A symmetric, damped-oscillatory "sinc" shape centered at $n = 0$. Peak of $1/3$ at $n = 0$, decaying envelope $\sim 1/(\pi n)$, with exact zero crossings at every third sample ($n = \pm 3, \pm 6, \dots$) because $\sin(n\pi/3) = 0$ there. The tails extend (in principle) to $n = \pm\infty$, oscillating in sign.

c) Why this can't be an exact real-time FIR filter

$h_d[n]$ is **two-sided** (nonzero for $n < 0$, so it's non-causal) and has **infinite duration** (it never exactly reaches zero for finite n). A real-time FIR filter must have a finite number of taps and can only use present and past inputs (causal). Since $h_d[n]$ is both infinite in length and non-causal, it cannot be implemented exactly — it must be truncated (finite length) and shifted (to be causal), which is the origin of the windowing/Gibbs phenomenon discussed in Problem 5.

d) Causal shift for $N = 61$

For an odd length N , truncate $h_d[n]$ over $n \in [-(N-1)/2, (N-1)/2] = [-30, 30]$, then shift right by $(N-1)/2 = 30$ samples:

$$h[n] = h_d[n - 30], \quad n = 0, 1, \dots, 60$$

Constant group delay = $(N - 1)/2 = 30$ samples.

Problem 5: Window Tradeoff (51-tap filters)

a) Narrower transition band

Filter A (rectangular window) has the narrower transition band — the rectangular window has the narrowest main lobe of common windows for a given length.

b) Lower stopband sidelobes

Filter B (Hamming window) has much lower stopband sidelobes, because the Hamming window's smooth tapering greatly suppresses sidelobe energy compared to the abrupt rectangular truncation.

c) Approximate first-sidelobe levels

- **Rectangular window:** first sidelobe ≈ -13 dB relative to the main lobe peak

- **Hamming window:** first sidelobe ≈ -41 to -43 dB relative to the main lobe peak (minimum stopband attenuation typically quoted around 53 dB)

d) Why longer rectangular windows don't remove Gibbs overshoot

Increasing the rectangular window length N narrows the transition band (compresses the ripple pattern closer to the discontinuity in frequency), but it does **not** reduce the *height* of the overshoot. This is the Gibbs phenomenon: near a discontinuity, truncating a Fourier series produces a fixed **$\sim 9\%$ overshoot** of the jump size regardless of how many terms are kept. Increasing N only moves the ripples closer to the discontinuity and increases their density — it does not make the peak overshoot amplitude shrink. To reduce the overshoot itself you need a different (tapered) window, such as Hamming, not just a longer rectangular one.

Problem 6: Symmetry and Linear Phase

Given: real FIR filter of length N with $h[n] = h[N - 1 - n]$, $n = 0, \dots, N - 1$.

a) Derivation

Let $M = (N - 1)/2$ (the center of symmetry). Factor out the linear-phase term:

$$H(e^{j\omega}) = \sum_{n=0}^{N-1} h[n] e^{-j\omega n} = e^{-j\omega M} \sum_{n=0}^{N-1} h[n] e^{-j\omega(n-M)}$$

Substitute $m = n - M$, so the sum runs over $m = -M, \dots, M$ (symmetric interval, since $N - 1 - M = M$). Define $g[m] = h[m + M]$. Because $h[n] = h[N - 1 - n]$, we get $g[m] = g[-m]$: the shifted sequence is **even**.

$$\sum_{m=-M}^M g[m] e^{-j\omega m} = g[0] + \sum_{m=1}^M g[m] (e^{-j\omega m} + e^{j\omega m}) = g[0] + 2 \sum_{m=1}^M g[m] \cos(\omega m)$$

The right-hand side is manifestly **real-valued** — call it $A(\omega)$. Therefore:

$$H(e^{j\omega}) = e^{-j\omega(N-1)/2} A(\omega), \quad A(\omega) \text{ real}$$

which is exactly the required form.

b) Group delay and relevance to ML feature extraction

The phase is $\theta(\omega) = -\omega(N - 1)/2$ wherever $A(\omega) \neq 0$ ($A(\omega)$ can go negative, contributing an extra constant π jump, but the *linear* term is unaffected). The group delay is:

$$\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega} = \frac{N - 1}{2} \quad (\text{constant, independent of } \omega)$$

Why this matters for ML feature extraction: a constant group delay means every frequency component of the signal is delayed by exactly the same number of samples — there is no phase distortion, so the filtered waveform is simply a delayed, undistorted copy of the shape that would result from an ideal (zero-delay) filter. In a feature-extraction pipeline (e.g., a filterbank producing multiple sub-band signals, as in MFCC-style audio features), this guarantees that transients, onsets, and relative timing relationships stay aligned across all filter channels — the fixed, known delay $(N-1)/2$ can simply be subtracted out. Nonlinear-phase filters would smear or shift different frequency components by different amounts, corrupting timing-sensitive features and complicating alignment across channels.

Problem 7: Pole Radius and Time-Domain Decay

For a causal system with a complex-conjugate pole pair $p = re^{\pm j\theta}$, a partial-fraction expansion of $H(z)$ contains terms of the form:

$$Ap^n u[n] + A^* p^{*n} u[n] = 2|A| r^n \cos(n\theta + \phi) u[n]$$

Why the decay is r^n : each pole contributes an impulse-response term proportional to $p^n = r^n e^{jn\theta}$. The magnitude of this term is $|p|^n = r^n$ exactly, since $|p| = r$. The angle θ only produces oscillation (a damped sinusoid); it's the magnitude r that controls the decay envelope.

Why $r \rightarrow 1$ increases ringing duration: the "time constant" of the decay behaves like $\tau \approx -1/\ln r$. As $r \rightarrow 1^-$, $\ln r \rightarrow 0^-$, so $\tau \rightarrow \infty$ — the envelope r^n decays more and more slowly, meaning the impulse response keeps oscillating (ringing) for many more samples before becoming negligible.

Why $r \rightarrow 1$ sharpens the frequency resonance: geometrically, $|H(e^{j\omega})|$ near the pole's angle is inversely proportional to the distance from the point $e^{j\omega}$ on the unit circle to the pole. That distance reaches its **minimum value of $(1 - r)$** exactly at $\omega = \theta$. As $r \rightarrow 1$, this minimum distance $(1 - r) \rightarrow 0$, so the reciprocal-distance factor blows up right at $\omega = \theta$, producing a tall, narrow resonance peak — i.e., a high-Q (high-quality-factor) filter.

Connecting the two effects: both are governed by the same distance $(1 - r)$ of the pole from the unit circle. A pole close to the unit circle is simultaneously (1) close in distance to the point $e^{j\theta}$, producing a sharp/narrow frequency peak, and (2) close to magnitude 1, producing very slow amplitude decay in time. This is a time-bandwidth duality: narrow bandwidth (sharp resonance) and long ringing time are two manifestations of the same underlying geometric fact — the pole sitting near the unit circle. Roughly, the resonance's 3-dB bandwidth $\Delta\omega \approx 2(1 - r)$ and the ringing time constant $\tau \approx 1/(1 - r)$, so $\Delta\omega \cdot \tau \approx \text{constant}$: you cannot make the resonance arbitrarily sharp without paying for it in longer time-domain ringing, and vice versa.